MoneyPrinter2

How the trading bot works, how to run it, and what the research found

In one paragraph. Three small programs ("bots"), one per market, watch EURUSD, GBPUSD and gold on your MetaTrader 5 demo account and trade on their own. Each bot runs several strategies and trades whichever one has a setup; two strategies on the same market can even hold trades at the same time. EURUSD and GBPUSD are traded around the London open (the *Sweep* and the *Breakout*) and with a daily *swing pullback*. Gold is traded with two daily trend-following systems (slow and fast). Each trade risks 1% of a $50,000 sizing base (about $500 CAD), and all bots together never risk more than 5% at once. Over 2021-2025 the combination backtested at **+69% total (about 11% a year), with a worst drop of -9.7% and no losing year**. That is a backtest, not a promise. The demo period is there to check it.

The three bots and their strategies

Market bot	Strategy	Chart	Trades / yr	When it trades (Toronto time)
EURUSD	London Sweep + London Breakout, daily trend filter	5 min	~27	3:00-6:00 AM entries, closed by 11:00 AM
	Daily swing pullback (buy dips in uptrends, 100-day trend)	1 hour	~10	Decides daily (7 PM), holds 1-10 days
GBPUSD	London Sweep + London Breakout (cross-checks EURUSD)	5 min	~38	3:00-6:00 AM entries, closed by 11:00 AM
	Daily swing pullback, 200-day trend	1 hour	~11	Decides daily, holds 1-10 days
XAUUSD	Gold trend following, 55-day, buy-only	1 hour	~5	Decides daily (7 PM), holds weeks
	Gold trend following, 20-day, buy-only	1 hour	~6	Same

Backtest results, 2021-2025 (1% risk per trade, your account's real costs)

	5-year return	Per year	Worst drop	Profit factor	Trades
EURUSD stack	+9.3%	1.8%	-11.1%	1.25	135
GBPUSD stack	+21.0%	3.9%	-8.5%	1.32	192
EURUSD pullback	+2.6%	0.5%	-2.9%	1.40	52
GBPUSD pullback	+3.6%	0.7%	-1.3%	1.87	54
Gold trend (55-day)	+14.9%	2.8%	-4.3%	3.55	24

	5-year return	Per year	Worst drop	Profit factor	Trades
Gold trend fast (20-day)	+18.1%	3.4%	-3.6%	2.94	32
All strategies together	**+69.4%**	**11.2%**	**-9.7%**	**1.54**	**489**

Year	2021	2022	2023	2024	2025
All strategies	+12.6%	+13.1%	+0.2%	+4.3%	+27.3%

The four newer strategies were chosen from many candidates, so some of their backtest strength is luck. Expect live results to be lower.

Profit factor = money won / money lost. Above 1.0 makes money; 1.3+ is solid for this kind of strategy. "Worst drop" is the largest fall from a peak.

1. A day in the life of the bot

All strategy times are London time, because that's when these markets are most active. Toronto is 5 hours behind London for most of the year. For a couple of weeks in March and in late Oct/Nov the gap is 4 hours, because the UK and North America change their clocks on different dates.

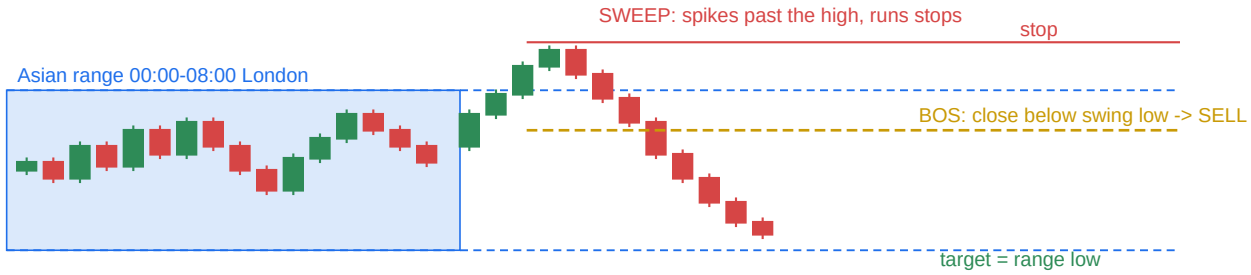
London	Toronto	What happens
00:00-08:00	7 PM-3 AM	EURUSD/GBPUSD bots record the overnight ("Asian") high and low: the **range**. The gold and pullback bots make their once-a-day decisions at 00:00 London.
08:00-10:30	3:00-5:30 AM	The **Sweep** watches for a fake break of the range followed by a sharp reversal.
10:30-11:00	5:30-6:00 AM	If no Sweep happened, the **Breakout** checks whether price broke out of the range and is still outside it.
13:30-16:00	8:30-11:00 AM	New York opens. This is where most of the profit on open trades is made.
16:00	11:00 AM	Any EURUSD/GBPUSD trade still open is closed. Gold trades are left open.
Every 5 min	Every 5 min	Each bot reads the newest closed candle, updates what it sees, writes its log, and redraws the chart overlay.

At most **one trade per pair per day** for EURUSD and GBPUSD. Mondays are skipped (Monday lost money in 4 of 5 years). Many days have no trade at all. That's normal: the bot only acts when its rules line up.

2. The strategies

2.1 London Sweep (a fake-out reversal)

Big players often push price just past an obvious level (the overnight high or low) to trigger other traders' stop-losses, then reverse. The Sweep tries to catch that reversal.



- **Sweep:** between 08:00 and 10:30, price pokes 1-15 pips past the range high (or low). Further than 15 pips counts as a real move, not a fake-out.
- **Confirmation (BOS, "break of structure"):** a candle closes back through the last small swing point made before the spike. The trade is taken against the spike.
- **SMT check:** on EURUSD the trade is only valid if GBPUSD did *not* make the same spike (and vice versa). Two pairs disagreeing is a sign of a genuine stop hunt.
- **Stop:** just beyond the spike's extreme (max 25 pips). **Target:** the other side of the range, and only if that's at least 1.5 times the risk.

- **Rare but strong:** only ~3 trades a year per pair, averaging about +0.6R each.

2.2 London Breakout (the workhorse)

When the overnight range is reasonably tight, the London open often breaks it decisively. About 90% of FX trades come from this strategy.

- **Range filter:** the overnight range must be 22-80 pips. Narrower ranges whipsaw; wider ones have usually already moved.
- **Patience:** it notes a break between 08:00 and 10:30 but waits until 10:30. If price is back inside the range by then, it was a false break and nothing is traded.
- **Stop:** the opposite side of the range (about 45 pips on average). **Target:** 2 times the risk. **Breakeven:** at +1R the stop moves to the entry price.
- **Most trades end at 16:00**, not at the stop or target, because the target is far away (see section 4).

2.3 The daily trend filter (looking at the bigger chart)

Before any EURUSD/GBPUSD trade, the bot checks the **50-day average** of daily closing prices. Price above it: only buys are allowed. Below it: only sells. This is the bot's version of a trader checking the daily chart before trading the 5-minute one. It cut trades by about 40% but roughly doubled profit per trade. Every average length from 30 to 200 days worked, so it isn't a lucky setting.

2.4 Gold daily trend following

Gold tends to move in long, strong trends. This strategy ignores the minute-to-minute noise and rides those trends.

- **Buy** when a daily close is higher than the highest price of the previous 55 days. Buy-only: selling gold lost money in every test.
- **Stop:** 2 times gold's average daily range below the entry, then it **trails** 2 times that range behind the best price reached. It only ever moves up.
- **Exit:** the trailing stop, or a daily close below the lowest price of the last 20 days.
- **Few trades, big wins:** ~5 a year, winners can be several R. Most of the backtest profit came in 2024-2025 (a strong gold rally). In sideways years it roughly breaks even.
- **Holds overnight,** so financing ("swap") applies. The backtest assumes a pessimistic -6.5% a year on the position's value.
- **Fast version (XAUUSD fast):** the same rules with a 20-day high to buy and a 10-day low to exit. It catches shorter gold trends the slow system misses. Both can be in a trade at the same time (2% at risk on gold).

2.5 Daily swing pullback (EURUSD pullback, GBPUSD pullback)

A completely different style from the London strategies: buy a short, sharp dip inside a bigger uptrend, and sell the bounce a few days later.

- **Trend:** the daily close versus its 100-day average (EURUSD) or 200-day average (GBPUSD).
- **Entry:** in an uptrend, buy when the 2-day RSI of daily closes drops below 10 (price fell hard for 1-3 days). In a downtrend, sell when it rises above 90.
- **Exit:** when the daily close gets back above the 5-day average (or below it for sells), or after 10 days. There's a protective stop at 3 times the daily range.
- **Why it's here:** about 70% of trades win, and it makes and loses money at different times from the London strategies (correlation about 0), so it smooths the overall results. Profits are small per pair (+0.5-0.7% a year) but steady.

3. Money management: how big each trade is

The bot doesn't pick a lot size directly. It decides **how much to lose if the stop is hit** and works the size backwards from the stop distance. A wide stop means a small position; a tight stop means a bigger one. The money at risk is the same either way.

```
risk per trade  = 1% x 50,000 (live.equity_cap)       = 500 CAD
position size   = 500 CAD / (stop distance x USD->CAD rate)

Example: GBPUSD breakout, stop 31 pips (0.0031), USD->CAD 1.42
         500 / (0.0031 x 1.42) = ~113,000 units = 1.13 lots
```

- **Your account is in CAD,** while EURUSD, GBPUSD and gold make profits in USD. The bot asks MT5 for the live conversion rate (about 1.42) so the risk really is 500 CAD.
- **Daily loss limit:** after losing 2% (1,000 CAD) in a day, no new trades open until tomorrow.
- **Leverage cap:** a position can never exceed 10 times the sizing base.

- **Portfolio cap:** before any trade, the bot adds up what all open bot trades would lose at their stops. If that plus the new trade is over 5% (2,500 CAD), the trade is skipped (logged as RISK CAP). Trades whose stop is already at breakeven count as zero.
- **Sizing base:** `equity_cap: 50000` in config.yaml sizes trades as if the account were exactly 50,000, plus or minus the bot's own profit and loss since it started.
- **At 1% risk** the backtest's worst drop was about -11% (-5,500 CAD). At 0.5% risk, returns and drops both roughly halve.

4. Exits: why winners and losers look similar

How EURUSD/GBPUSD trades end	Share	Typical result
Closed at 16:00 London	69%	small, either way (average +0.26R)
Stop loss hit	25%	about -0.9R
2R target hit	6%	about +2R

The Breakout's stop sits across the whole overnight range, so a 2R target needs a move most afternoons don't make. Most trades end at 16:00 wherever price is, which makes wins and losses similar in size (average win +0.79R, average loss -0.59R). The edge is being on the right side slightly more often and slightly bigger: about +0.1R per trade.

Every alternative exit was tested and did worse: closer or further targets, earlier or no breakeven, exiting at 14:00 or 15:00 (cuts profit to +3% / +19%), tighter stops (profit fell to 0%), and 9 trailing-stop versions. The profit is made during the New York open, so the trade has to stay open until 16:00. *R* = the amount risked on a trade; +2R means the trade made twice what it risked.

5. Running the bots

5.1 Start, stop, check

To...	Do this
Start everything	Double-click `start_bots.bat`. It opens MT5 if needed, then three windows: fxbot EURUSD / GBPUSD / XAUUSD. Each market bot runs all of that market's strategies, and restarts itself 60 s after a crash.
Stop the bots	Double-click `stop_bots.bat` (or close the windows). Open trades stay open, with their stop and target held at the broker.
Run one bot by hand	`venv\Scripts\activate`, then `python run_live.py --pair EURUSD` (runs all EURUSD strategies)
Watch without trading	add `--dry-run` (logs what it would do, sends nothing)
See how the demo is doing	`python run_report.py`: every bot trade, P/L in CAD, R, slippage, and the backtest's expectation for comparison

A second copy of the same pair refuses to start ("Another run_live.py is already running"), so starting the bots twice can't double your trades.

5.2 Keeping it running 24/7

- **Auto-start at login** (run once in PowerShell):
  `schtasks /Create /TN "MoneyPrinter2 bots" /TR "C:\Users\harsh\MoneyPrinter2\start_bots.bat" /SC ONLOGON /RL LIMITED /F`
- **No sleep on power**: `powercfg /change standby-timeout-ac 0`
- **MT5 must stay open and logged in.** Tick "Save password" when logging in, and keep the *Algo Trading* button green.
- **Windows Update** can restart the PC overnight. Set Active Hours, or check in the morning that the three windows are back.

5.3 Seeing what the bot "thinks"

Terminal windows. Each market's window explains itself in plain English, like code comments, e.g. # *London wants to SELL GBPUSD: price broke below the overnight range and stayed there.* It covers startup, what each strategy is waiting for, every order, stop move, close (with P/L), skipped trades and why, and a daily summary. Add `--verbose` to also see the technical lines.

Journal. logs/journal_<date>.log collects the important events from all three markets in one file. The FxbotOverlay indicator also prints them in MT5's **Experts** tab.

Chart overlay. In MT5, open a 5-minute EURUSD., GBPUSD. or XAUUSD chart, then Navigator (Ctrl+N), Indicators, **FxbotOverlay**, and drag it onto the chart. The bot redraws it after every candle. It only draws; it never trades.

On the chart	Meaning
Blue box / blue lines	the overnight range (Asian high and low)
Grey dotted lines	the sweep limit: past this it's a real move, not a fake-out
Red line	where price swept (the spike's extreme)
Gold dashed line	the trigger: a close past it means the bot enters
Green dashed lines	breakout buy and sell zones
Silver dotted line	the 50-day average (the daily trend)
Gold chart: green/grey	55-day high (buy trigger) and 20-day low (exit)
Top-left text	status, e.g. "high swept -> waiting for close below 1.12307 (9 bars left)"

Logs are in `logs\live_<PAIR>_<date>.log`. The useful lines:

```
STATUS Sweep: trend DOWN -> sells only; high swept -> waiting for close below
1.12307
ORDER SELL 0.14 lots @ ~1.34283  SL=1.34595  TP=1.33644  spread=2.0 pips
Order filled: ticket=14427020 fill_price=1.34283 requested=1.34283
Stop moved to 1.34283
CLOSED short_break SELL ... P/L 125.99  reason=[tp 1.33644]
```

6. What was tested (and why most of it isn't used)

Each idea was backtested on 5 years of 1-minute data with your account's real costs (EURUSD 1.4 pips, GBPUSD 2.0, gold $0.80), and judged year by year. A strategy that made money only in one lucky year doesn't count.

Idea	Result	Verdict
London Sweep + Breakout with trend filter (EUR, GBP)	+30% at 1% risk, PF 1.30	**LIVE**
Gold daily trend following, buy-only (55-day and 20-day)	+14.9% / +18.1%, PF 3.55 / 2.94	**LIVE**
Daily swing pullback on EURUSD / GBPUSD	+2.6% / +3.6%, PF 1.40 / 1.87, ~70% winners	**LIVE**
Volatility breakout, previous-day breakout	lost on all three markets	rejected
NR7 / NR4 / inside-day breakouts	about break-even (gold inside-day +12% but raised drawdown)	rejected
Same London strategies on gold or AUDUSD	about break-even or losing	rejected
Opening-range breakouts (London / NY open)	lost 27-91% in 5 years	rejected
Pullback scalping (RSI-2 on 5m / 15m / 1h)	lost 21-99%, 190-780 trades/yr	rejected
Support/resistance lines ("key levels")	all 48 versions lost money	rejected
London-close reversal / continuation	both directions lost	rejected
Daily trend following on EURUSD / GBPUSD	about zero, even with no swap	rejected
1-minute candles instead of 5	worse (stops hit by noise)	rejected
Trading Mondays / longer windows / NY session sweep	more trades, less profit	rejected

About bots that make 4+ trades a day. The fast strategies above trade that often and lose, even at the cheaper costs of a raw-spread account. Every trade pays the spread: at 1.4 pips, 1,000 trades a year cost 1,400 pips before any edge. Bots advertising many trades and smooth profits are often grid/martingale systems (add to losers; they look great until one trend wipes the account) or backtests run without costs.

Robustness checks on the live setup

- **Not overfit to the past:** settings chosen on 2021-2023 (+7.5%/yr) also worked on 2024-2025 (+8.5%/yr), data they were never tuned on.
- **Survives higher costs:** still profitable at a 2-pip spread on both pairs.
- **Stable settings:** nearby values of the key settings (range 20-25 pips, trend 30-200 days, gold 40-100-day channel) all made money. One setting that only worked at an exact value would be a red flag.
- **Honest weak spots:** EURUSD alone roughly broke even in 2024-2025. Gold's profit depends on gold trending. Gold's real overnight financing is an estimate, since the demo shows 0.

7. The demo plan

Step	What
Now	Run all three bots on the demo 24/7. Don't change strategy settings during the test.
Weekly	`python run_report.py`. Compare live win rate, average R and trades per week with the backtest.
~2 months (30+ trades)	Decide: **matches the backtest**, so keep going (consider a raw-spread account or a real account at low risk). **Much worse**, so find out why (costs, execution, market change) before changing anything. **Unclear**, so keep running; small samples swing a lot.

What to expect: roughly 2 trades a week across all strategies, a gold trade every month or two, losing weeks and possibly losing months. One trade, or one week, tells you almost nothing.

8. Reference

Files

File	What it is
config.yaml	every setting: risk, costs, strategies, the three markets (pairs:), live options
run_live.py	the live/demo bot (one per market: --pair EURUSD | GBPUSD | XAUUSD)
start_bots.bat / stop_bots.bat / run_forever.bat	24/7 launchers
run_report.py	demo results vs backtest expectations (writes results/live_trades.csv)
run_validate.py	walk-forward, per-year and robustness checks of the live setup
run_arsenal.py	tests every strategy on every market (the research table in section 6)
fxbot/strategies/	the strategy rules (london_sweep, london_breakout, trend_follow, ...)
fxbot/brokers/mt5_broker.py	talks to MetaTrader 5 (orders, stops, CAD conversion, gold contract size)
fxbot/brokers/sim_broker.py	the backtest simulator (spread, slippage, swap, gaps)
mt5/FxbotOverlay.mq5	the chart-drawing indicator for MT5
logs/	one log per bot per day

Settings you might change

Setting (config.yaml)	Now	Effect
risk.risk_per_trade_pct	1.0	% of the sizing base lost if a stop is hit. Halve it to halve both returns and drawdowns.
live.equity_cap	50000	sizing base. null = use the real account equity.
live.max_open_risk_pct	5.0	most that all open bot trades together may lose at their stops
risk.max_daily_loss_pct	2.0	no new trades for the day after this loss
pairs: spread_pips	1.4 / 2.0 / 8.0	cost assumption for backtests (live uses the real spread)
live.allow_live	false	must be true, plus --allow-live, before it will touch a real-money account

Safety built in

- Refuses real-money accounts unless you explicitly enable it twice (config + command flag).
- Every order is sent with its stop loss attached, so a crash or power cut still leaves the stop at the broker.
- One bot per market (lock file). Each strategy labels its trades with its own magic number: EURUSD London 12345 / pullback 12445, GBPUSD London 12346 / pullback 12446, gold 55-day 12347 / 20-day 12447. Strategies only manage their own trades, never yours or each other's.
- Closes positions by ticket, which is required on your hedging-mode account. Without this, a close would open an opposite trade instead.
- Daily loss limit, leverage cap, and the CAD conversion so the risk is what it says.

- Spread guard: no new entry while the live spread is over 2.5 times normal (news spikes, rollover).
- Drawdown circuit breaker: no new trades once the account is 15% below its best level. Open trades keep their stops. Delete data/state/peak_equity.json to resume after checking why.
- Stall watchdog: if the market is open but MT5 stops sending candles, the bot reconnects.
- **News filter:** no new trades 30 minutes either side of high-impact news for that market's currencies (EURUSD: USD and EUR, GBPUSD: USD and GBP, gold: USD). The bot reads the Forex Factory weekly calendar every 4 hours and caches it. If the download fails it keeps the last copy. Open trades are not closed. Skips are logged as NEWS: skipping entry. Backtest cost: 69.4% to 69.1% over 2021-2025, same drawdown.

Nothing here is financial advice. Backtests describe the past under assumptions; live results will differ. Only trade money you can afford to lose.